深度學習實作與應用
Deep learning and its applications

Project Announcement &
Data processing

IM5062, Spring 2026

Some slides from Qingqing Huang, Mu Li, Alex Smola

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Week	Date	Topic	備忘
1	2月24日	Course introduction	
2	3月3日	Basic Neural Network (I): from regression to neural networks	
3	3月10日	Basic Neural Network (II): forward propagation	
4	3月17日	Basic Neural Network (III): backward propagation Pytorch Tutorial	HW1 announce
5	3月24日	Basic Neural Network (VI): optimizers, normalization, regularization, evaluation Static Embedding Techniques	
6	3月31日	Convolutional Networks: Architectures, convolution / pooling layers	HW2 announce
7	4月7日	Recurrent Neural Networks: RNN, GRU, LSTM, Seq2Seq model	
8	4月14日	Attention mechanism, Transformer	4/13 寄分組名單
9	4月21日	Midterm	
10	4月28日	Project Proposal	Report 1
11	5月5日	Guest Lecturer: 國泰銀行 顏勝洲經理	Report 2
12	5月12日	BERT, GPT, LLMs	HW3 announce
13	5月19日	Guest Lecturer: 中研院資訊所王建堯副研究員	Report 3
14	5月26日	Emerging topics	
15	6月2日	Project presentation	Report 4
16	6月9日	Project presentation	Report 5

Final project: 30%

- 4 people in a team
- Novel problem: 5%
- Data collection and validation: 5%
- Novel approach: 5%
- Results w/o comparisons: 5% (w/ 5%)
- Proposal/Presentation/Report: 5%

Proposal

- Task definition:
  - Input
  - Output
- Data collection
  - 如果使用公開的資料集，須說明來源、細節與比較對象
  - 自行蒐集的資料及：
    - 整體 Data數量與分布 (分別說明input和output)
    - 如何切 training/validation/testing
  - Data validation & Preprocessing
- Approach:
  - 預計使用的方法與緣由
- Evaluation:
  - Metrics
  - 預計比較的對象

Final report

- Introduction
- Method
  - Task Definition
  - Approach
- Data collection and validation
- Evaluation
  - Metrics
  - Baselines
  - Results (Numerical & Case Study)
- Conclusion

Deadline

- 4/13 中午前，請組長寄信給助教分組(4人)名單
- 4/27 中午前公告報告順序
- 4/28 Proposal:
  - 不須繳交檔案
  - 報告15分鐘
- 6/9 Project presentation:
  - 報告25分鐘，當天提供4頁一面的投影片給老師
  - 6/9 5pm 繳交投影片、5 page 書面資料 (不包含reference與工作分配)
    - 格式：word和tex檔 、pdf檔

Open-source datasets

- Popular datasets
  - MNIST: digits written by employees of the US Census Bureau
  - ImageNet: millions of images from image search engines
- More Image, Text, Audio, etc. data at https://en.wikipedia.org/wiki/List_of_datasets_for_machine-learning_research
- Search
  - Kaggle: https://www.kaggle.com/datasets
  - Google dataset search: https://datasetsearch.research.google.com/

Dataset comparison

	Pros	Cons
Academic datasets	Clean, proper difficulty	Limited choices, too simplified, usually small scale
Competition datasets	Closer to real ML applications	Still simplified, and only available for hot topics
Raw Data	Great flexibility	Needs a lot of effort to process

Make dataset on your own

- Web crawling VS scrapping
  - Crawling: indexing whole pages on Internet
  - Scraping: scraping particular data from web pages of a website

- Legal Considerations
  - Web scraping isn’t illegal by itself
  - But you should
    - NOT scrape data have sensitive information (E.g. private data involving username/password, personal health/medical information)
    - NOT scape copyrighted data (E.g. YouTube videos, Flickr photos)
    - Follow the Terms of Service that explicitly prohibits web scraping

Labelling

- Enough data/label
  - Notice the data distribution

- Not Enough data/label
  - Crowdsourcing: leverage global labelers to manually label data
    - Quality Control:
      - Sending the same task to **multiple labelers**, then determine the label by **majority voting**
      - Improve: prune low-quality labelers

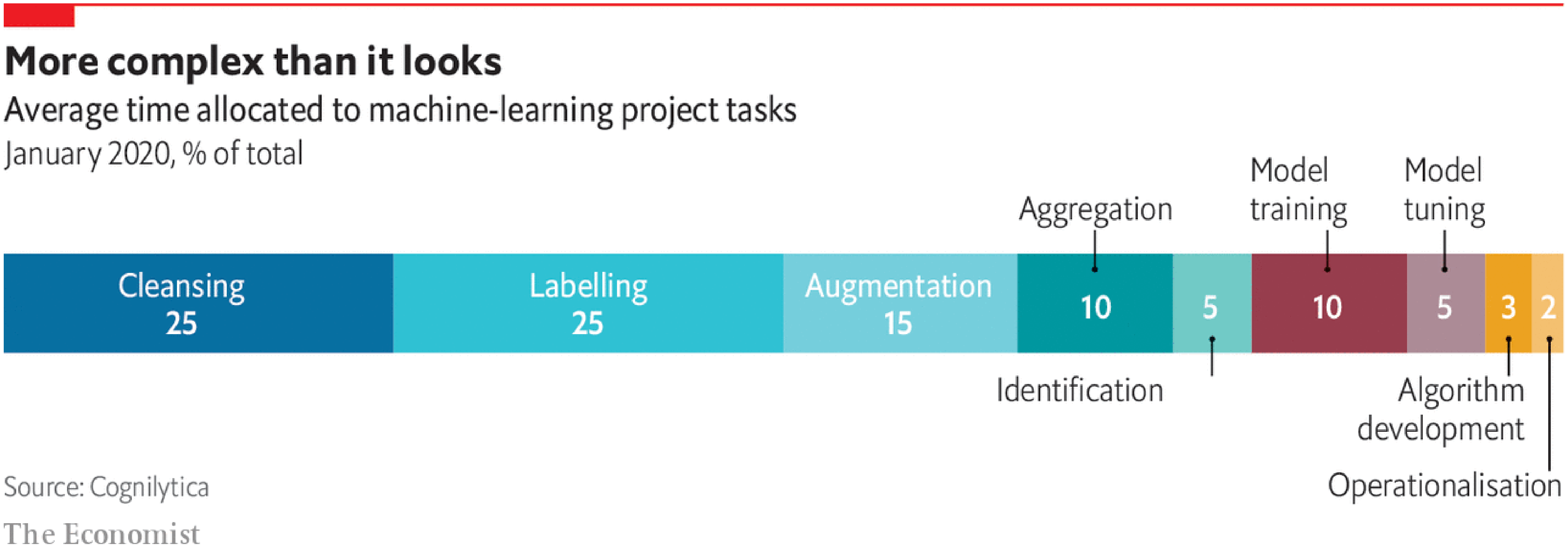
  - Data programming: heuristic programs to assign noisy labels
    - Domain specific heuristics to assign labels
    - Keyword search, pattern matching, third-party models

Data cleaning

- Outliers: data values that significantly deviate from other observations
- Rule violations: data values violate integrity constraints such as “Not Null” and “Must be unique” and “Non negative”
- Pattern violations: data values violate syntactic and semantic constraints such as formatting, misspelling

Data transformation

- Normalization:
  - Real value features:
    - Min-max
    - Z-score
    - Log scaling
  - Text:
    - Stemming
    - Lemmatization
    - Tokenization



Roh, Yuji, Geon Heo, and Steven Euijong Whang. "A survey on data collection for machine learning: a big data-ai integration perspective." *IEEE Transactions on Knowledge and Data Engineering* (2019).